

B4.

$$\int \frac{x^2+4}{(x+2)^2} dx = \int \frac{x^2+4}{x^2+4x+4} dx$$

B4

$$\frac{x^2+4}{-x^2-4x-4} \quad \frac{x^2+4x+4}{1}$$

$$I = \int \left(1 - \frac{4x}{(x+2)^2} \right) dx.$$

$$\frac{4x}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2}$$

$$\frac{4x}{(x+2)^2} = \frac{A(x+2) + B}{(x+2)^2}$$

$$4x = A(x+2) + B.$$

$$x = -2 \quad \boxed{-8 = B}$$

$$x = 0 \quad 0 = 2A - 8 \rightarrow \boxed{A = 4}$$

$$I = \int \left[1 - \left(\frac{4}{x+2} - \frac{8}{(x+2)^2} \right) \right] dx =$$

$$= \int \left(1 - \frac{4}{x+2} + \frac{8}{(x+2)^2} \right) dx =$$

$$= x - 4 \ln|x+2| + 8 \frac{(x+2)^{-2+1}}{-2+1} + k =$$

$$\boxed{I = x - 4 \ln|x+2| - \frac{8}{x+2} + k.}$$

B4 II) $\int (x+2) \sin(3x) dx$

B4

Zatukoko metodoojot :

$$\int u(x) dv = u(x) \cdot v(x) - \int v(x) du.$$

$$\left\{ \begin{array}{l} u = x+2 \rightarrow du = dx \\ dv = \sin 3x dx \rightarrow v = \frac{1}{-3} \int -3 \sin 3x dx \end{array} \right.$$

$$v = -\frac{1}{3} \cos(3x)$$

$\cos 3x \rightarrow -3 \sin 3x$

$$I = (x+2) \left(-\frac{1}{3} \right) \cos 3x - \int -\frac{1}{3} \cos(3x) dx =$$

$$= -\frac{x+2}{3} \cos(3x) + \frac{1}{3} \int \frac{3}{3} \cos(3x) dx =$$

$\sin 3x \rightarrow 3 \cos 3x$

$$I = \frac{-x-2}{3} \cos(3x) + \frac{\sin(3x)}{9} + K.$$